

Principles of Observation and Mechanical Observation

From Superposition to Retained Reality through Observer Architecture

A Unified Mechanics working paper

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July 2026

Central thesis

Superposition is the coherent retention of unresolved admissible actions. Observation is the physical production of a retained distinction relative to another system. Mechanical observation is the ordered observer architecture that converts those unresolved actions into stable, discriminable records. Human scientific observation adds reflexive modelling and semantic denotation of the machinery itself.

Worked physical system: a complete Stern-Gerlach measurement chain

Abstract

This paper introduces the Principle of Observation (PO) as a universal operational principle and the Principle of Mechanical Observation (PMO) as its structured physical realisation. The distinction resolves an ambiguity in the term observer. At the minimal level, observation is any interaction through which a distinction in one system becomes retained relative to another. Mechanical observation adds an internal architecture that selects a measurable distinction, couples it to distinguishable channels, propagates and stabilises the resulting signal, and closes it into a record. Human scientific observation is a reflexive mechanical subclass capable of modelling and semantically denoting both the target and the machinery of observation. The framework is connected mathematically to quantum superposition, unitary system-apparatus coupling, decoherence, positive-operator-valued measures, and quantum instruments. A full Stern-Gerlach experiment is then decomposed as a real physical example, from the incoming spin superposition through magnetic separation, detector registration, environmental record stabilisation, and human interpretation. The result is a proposed operational bridge from unresolved quantum action to retained observer-relative reality.

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1. The Missing Stage after Superposition

The prior action-based reconstruction identified a quantum state as a normalised relational account of unresolved admissible logical actions. In a basis of mutually distinguishable alternatives,

$$|\psi_S\rangle = \sum_i c_i |s_i\rangle, \quad \sum_i |c_i|^2 = 1. \quad (1)$$

Equation (1) preserves every admissible alternative and the phase relations between them. It therefore describes an unresolved physical account, not an observed record. The next question is not merely how a probability is computed, but how an unresolved action-space becomes a persistent distinction in another physical system.

Operational hinge

Superposition specifies what must remain jointly retained before resolution. Observation specifies how one system becomes a retained account of another. Mechanical observation specifies the machinery that performs this conversion.

The complete bridge proposed here is

$$\text{Admissibility} \rightarrow \text{Superposition} \rightarrow \text{Mechanical distinction} \rightarrow \text{Correlation} \rightarrow \text{Propagation} \rightarrow \\ \text{Stabilisation} \rightarrow \text{Retained record} \rightarrow \text{Semantic closure.} \quad (2)$$

2. Principle of Observation

Principle of Observation (PO)

An observation occurs whenever a distinction in one system becomes physically related to and persistently retained by another system. Consciousness is not required at this level. Observation is the production of an observer-relative fact through retained physical correlation.

$$\mathfrak{D}_{\{O \leftarrow S\}} = \mathcal{C}_O \circ \mathcal{P}_O \circ \mathcal{R}_{\{OS\}} \circ \mathcal{D}_O(S).$$

Here S denotes the observed system and O denotes the observer or reference system. The components are

Operation	Mathematical role	Physical meaning
$\mathcal{D}_O(S)$	Distinction	A target-relative difference becomes available to O .
$\mathcal{R}_{OS}$	Relation	S and O physically interact or share a preparation correlation.
$\mathcal{P}_O$	Persistence	The difference leaves a stable enough change in O to matter later.
$\mathcal{C}_O$	Closure	The retained difference becomes an attributable observer-relative fact.

At this level, an atom can observe another atom in the relational sense: an interaction creates a correlation that changes the future behaviour of both. This agrees with relational approaches in which physical facts are relative to systems and no fundamental human observer-observed division is imposed.^[1]

2.1 Minimal relational observation

$$\Sigma_i c_i |s_i\rangle |o_0\rangle \rightarrow \Sigma_i c_i |s_i\rangle |o_i\rangle. \quad (3)$$

The observer states $|o_i\rangle$ now contain distinctions correlated with the alternatives $|s_i\rangle$. Equation (3) is already observation under PO because the relation has changed what is physically attributable relative to O . It is not yet necessarily a macroscopic measurement, a classical record, or a conscious experience.

3. Principle of Mechanical Observation

Principle of Mechanical Observation (PMO)

A mechanical observation is an ordered physical process in which the observer has an internal architecture that selects admissible distinctions, couples them to distinguishable channels, propagates and stabilises the resulting signal, and retains it as an identifiable record. Mechanical means implemented by a mechanism, not necessarily artificial.

$$\mathfrak{D}^{\{\text{mech}\}}_{\{O \leftarrow S\}} = \mathcal{C}_O \circ \mathcal{R}_{ec_O} \circ \mathcal{A}_O \circ \mathcal{P}roj_O \circ \mathcal{U}_{\{SM\}}.$$

The observer machinery is represented as the tuple

$$\mathcal{M}_O = (\mathcal{H}_M, \sigma_O, \mathcal{U}_{\{SM\}}, \{\Pi_k\}, \mathcal{A}, \mathcal{R}ec). \quad (4)$$

Its structure is

- $\mathcal{H}_M$: the state-space of the observing mechanism.
- σ_0 : its prepared ready state.
- U_{SM} : the system-machinery coupling.
- $\{\Pi_k\}$: the distinguishable output or pointer channels.
- $\mathcal{A}$: amplification, redundancy, or stabilisation.
- $\mathcal{R}ec$: production of a persistent record.

Subclass relation

Every mechanical observation is an observation, but not every observation is mechanical in the structured sense: $PMO \Rightarrow PO$, while PO does not imply PMO .

3.1 Why the structure is genuinely different

A minimal observer only needs to retain a correlation. A mechanical observer must impose an ordered channel architecture. It therefore carries additional internal distinctions: ready state versus triggered state, one output channel versus another, signal versus noise, transient response versus persistent record, and record versus interpretation.

$PO: S \leftrightarrow O \rightarrow \text{retained difference.}$ $PMO: S \rightarrow \text{selector} \rightarrow \text{transducer} \rightarrow \text{channel} \rightarrow \text{amplifier} \rightarrow \text{record.} \quad (5)$

4. Observer Architecture and Measurement Choice

The machinery is not a passive window onto a fully prewritten classical property. It defines the physical question enacted on the system. In general quantum measurement theory, a measurement is represented by effects $\{E_k^{\mathcal{M}}\}$ satisfying

$$E_k^{\mathcal{M}} \geq 0, \quad \sum_k E_k^{\mathcal{M}} = I, \quad (6)$$

$$p(k | \rho, \mathcal{M}) = \text{Tr}(\rho E_k^{\mathcal{M}}). \quad (7)$$

The same incoming state ρ can therefore yield different probability distributions under different observer architectures $\mathcal{M}$. The apparatus does not create arbitrary outcomes: the state and the interaction constrain them. But the apparatus determines which distinction is made mechanically available as a record.

4.1 Qubit example

For a spin-1/2 state with Bloch vector r , a measurement along a unit direction n has effects

$$\Pi_{\pm}^{\mathcal{M}}(n) = \frac{1}{2}(I \pm n \cdot \sigma), \quad (8)$$

$$p_{\pm}(n) = \text{Tr}(\rho \Pi_{\pm}^{\mathcal{M}}(n)) = \frac{1}{2}(1 \pm r \cdot n). \quad (9)$$

Rotating the measuring magnet changes n and therefore changes the distinction being physically enacted. Observer architecture is thus part of the complete state-to-record operation.

5. Superposition Transferred into Machinery

Let the system enter the observation in a coherent superposition and let the apparatus begin in a ready state $|M_0\rangle$:

$$|\psi_S\rangle|M_0\rangle = (\sum_i c_i |s_i\rangle)|M_0\rangle. \quad (10)$$

An ideal channel-defining interaction satisfies

$$U_{\{SM\}}(|s_i\rangle|M_0\rangle) = |s_i\rangle|M_i\rangle. \quad (11)$$

By linearity,

$$U_{\{SM\}}[(\sum_i c_i |s_i\rangle)|M_0\rangle] = \sum_i c_i |s_i\rangle|M_i\rangle. \quad (12)$$

Transfer proposition

The first stage of observation does not erase superposition. It transfers the unresolved action from the target alone into a larger correlated system-apparatus structure.

This is the exact transition from superposition to observer relation. Before coupling, the alternatives are carried by S. After coupling, they are carried by correlations between S and M. The observer machinery has made the alternatives mechanically discriminable, but it has not yet by unitary evolution alone produced one uniquely selected global branch.

6. Decoherence and Retained Records

Real apparatus records interact with an environment E. If the environment begins in $|E_0\rangle$ and becomes correlated with the pointer alternatives,

$$\sum_i c_i |s_i\rangle|M_i\rangle|E_0\rangle \rightarrow \sum_i c_i |s_i\rangle|M_i\rangle|E_i\rangle. \quad (13)$$

The reduced system-apparatus density operator is

$$\rho_{\{SM\}} = \sum_{\{i,j\}} c_i c_j^* \gamma_{\{ij\}} |s_i M_i\rangle\langle s_j M_j|, \quad \gamma_{\{ij\}} = \langle E_j | E_i \rangle. \quad (14)$$

When distinct records leave nearly orthogonal environmental traces, $\gamma_{ij} \approx 0$ for $i \neq j$, and locally

$$\rho_{\{SM\}} \approx \sum_i |c_i|^2 |s_i M_i\rangle\langle s_i M_i|. \quad (15)$$

Environment-induced decoherence explains why certain pointer records are stable and why interference between macroscopically distinct records becomes locally inaccessible.^[2]

Record stability condition

A candidate record is mechanically real for an observer when it remains distinguishable and recoverable through the subsequent interactions that constitute the observer's memory or readout chain.

$$\text{Record}(k) \Leftrightarrow \text{Dist}(M_k, M_j) \text{ persists under the relevant future dynamics for all } j \neq k.$$

This formulation deliberately separates stable record production from any one metaphysical interpretation of global outcome selection. Decoherence provides the mechanical stability and effective noninterference of records; interpretations of quantum mechanics disagree about the global ontology of the remaining branches. PMO identifies the common physical operation that every interpretation must still represent.

7. Quantum Instruments as Mechanical-Observation Maps

The standard mathematical object closest to PMO is a quantum instrument $\{\mathcal{J}_k\}$. It encodes both the probability of each record and the state transformation conditioned on that record:

$$p_k = \text{Tr}[\mathcal{J}_k(\rho)], \quad (16)$$

$$\rho_{\{S|k\}} = \mathcal{J}_k(\rho) / p_k, \quad (17)$$

$$\sum_k \text{Tr}[\mathcal{J}_k(\rho)] = 1. \quad (18)$$

Completely positive instruments can be represented through indirect measurement models involving a physical apparatus, a system-apparatus interaction, and an apparatus readout. This is precisely the mathematical content required by mechanical observation: an abstract state-to-outcome map must be physically realisable as machinery.^[3]

7.1 Explicit apparatus dilation

For apparatus state σ_0 , coupling U_{SM} , and pointer projector Π_k , an idealised instrument can be written

$$\mathcal{I}_k(\rho) = \text{Tr}_M[(I \otimes \Pi_k) U_{\{SM\}}(\rho \otimes \sigma_0) U_{\{SM\}}^\dagger (I \otimes \Pi_k)]. \quad (19)$$

Equation (19) is a complete mechanical-observation sentence in conventional mathematics: prepare machinery, couple system and machinery, distinguish pointer channel k , discard inaccessible apparatus detail, and retain the conditional system state.

8. Human Reflexive Observation

A human is not required for PO or for the mechanical production of a record. The specifically human scientific distinction appears when the observer can model the target, model the apparatus, re-enter the observation as a new object of observation, and communicate the result semantically.

$$\mathcal{O}_H^{\wedge(2)}(S) = \Lambda_H \circ \text{Rec}_H \circ \mathcal{O}_{\{H \leftarrow S\}^{\text{mech}}}. \quad (20)$$

Here Rec_H denotes reflexive re-entry and Λ_H denotes semantic denotation into language, diagrams, or mathematics. The human observer can ask why a basis was selected, what the detector excluded, how the signal was amplified, what assumptions entered the reconstruction, and whether a different observer architecture would yield another legitimate distinction.

Reflexive mechanical observer

A mechanical observer whose internal records can themselves become targets of further observation, comparison, revision, and semantic denotation.

Human scientific observation = PMO + recursive self-model + semantic closure.

The universe therefore contains observation without requiring a universal consciousness, while human consciousness and scientific practice realise a higher-order observer structure. The universe physically registers distinctions; humans can make the machinery of registration itself into a new distinction.

9. Worked Real-World System: Stern-Gerlach Measurement

A Stern-Gerlach apparatus provides a concrete physical system in which every stage of PO and PMO can be identified. A beam of neutral atoms with a magnetic moment passes through a nonuniform magnetic field. The spin degree of freedom becomes correlated with spatial motion, and a downstream detector converts the separated wavepackets into stable records. The experiment is routinely used as the canonical model of spin measurement. Modern analyses treat spin and position quantum-mechanically and explicitly study the role of apparatus resolution and environment.^[4]

9.1 Physical apparatus

Component	Observer role
1. Atomic source	Produces a beam of atoms and establishes the preparation context.
2. Collimator	Restricts transverse position and momentum so later splitting can be resolved.
3. Inhomogeneous magnet	Selects a spatial axis and couples spin to motion.

Component	Observer role
4. Free-flight region	Allows the correlated momentum alternatives to separate in position.
5. Detection screen	Transduces atom arrival into local material change.
6. Camera/electronics	Amplifies and stores the spot location as a durable record.
7. Human observer	Models the apparatus, identifies the selected basis, and denotes the result.

9.2 Incoming superposition

Choose the magnet's gradient direction as z . An incoming atom may be prepared in

$$|\psi_{\text{spin}}\rangle = \alpha|\uparrow_z\rangle + \beta|\downarrow_z\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (21)$$

Let $|\phi_0\rangle$ be the initial spatial wavepacket. The complete incoming state is

$$|\Psi_0\rangle = (\alpha|\uparrow_z\rangle + \beta|\downarrow_z\rangle) \otimes |\phi_0\rangle. \quad (22)$$

At this stage, the spin alternatives are admissible and coherently retained. There is not yet a z -record.

9.3 Mechanical distinction selected by the magnet

The magnetic interaction is approximately

$$H_{\text{int}} = -\mu \cdot B(r). \quad (23)$$

For a field with z -gradient $B_z(z) \approx B_0 + bz$, the two z -spin components experience opposite forces. Over an interaction time τ , the apparatus produces opposite momentum shifts $\pm \Delta p_z$. The unitary interaction therefore acts as

$$|\uparrow_z\rangle|\phi_0\rangle \rightarrow |\uparrow_z\rangle|\phi_+\rangle, \quad |\downarrow_z\rangle|\phi_0\rangle \rightarrow |\downarrow_z\rangle|\phi_-\rangle. \quad (24)$$

By linearity,

$$|\Psi_1\rangle = \alpha|\uparrow_z\rangle|\phi_+\rangle + \beta|\downarrow_z\rangle|\phi_-\rangle. \quad (25)$$

The magnet has not passively revealed a basis-independent label. Its physical orientation selected the z distinction and converted it into spatially separable channels. Rotating the magnet changes the projectors and therefore changes the mechanical observation performed.

9.4 Propagation and spatial resolution

During free flight, the wavepackets separate. A useful discriminability condition is

$$|\langle\phi_-|\phi_+\rangle| \ll 1. \quad (26)$$

When the overlap remains large, the machinery cannot reliably distinguish the alternatives. When it is small, the apparatus has converted spin distinction into a resolvable position distinction. The collimator, field gradient, transit time, atom mass, detector distance, and detector resolution all belong to the observer machinery because all determine whether equation (26) is achieved.

9.5 Detection and record formation

Let $|D_0\rangle$ be the ready state of the detector and $|D_+\rangle, |D_-\rangle$ be distinct detector records. Detection gives

$$\alpha|\uparrow_z\rangle|\varphi_+\rangle|D_0\rangle + \beta|\downarrow_z\rangle|\varphi_-\rangle|D_0\rangle \quad (27a)$$

$$\rightarrow \alpha|\uparrow_z\rangle|\varphi_+\rangle|D_+\rangle + \beta|\downarrow_z\rangle|\varphi_-\rangle|D_-\rangle. \quad (27b)$$

The detector then couples to many uncontrolled environmental degrees of freedom: emitted photons, lattice vibrations, electrical currents, memory states, and surrounding matter. Denote the resulting environmental states by $|E_+\rangle$ and $|E_-\rangle$:

$$|\Psi_2\rangle = \alpha|\uparrow_z, \varphi_+, D_+, E_+\rangle + \beta|\downarrow_z, \varphi_-, D_-, E_-\rangle. \quad (28)$$

The local coherence between the two detector records is multiplied by

$$\gamma = \langle E_- | E_+ \rangle. \quad (29)$$

For macroscopically distinct records, $|\gamma|$ becomes extremely small. The reduced detector description is then approximately

$$\rho_D \approx |\alpha|^2 |D_+\rangle\langle D_+| + |\beta|^2 |D_-\rangle\langle D_-|. \quad (30)$$

9.6 Quantum-instrument description

For an ideal z measurement, the effects are

$$\Pi_+ = |\uparrow_z\rangle\langle\uparrow_z|, \quad \Pi_- = |\downarrow_z\rangle\langle\downarrow_z|. \quad (31)$$

$$p_+ = \text{Tr}(\rho\Pi_+) = |\alpha|^2, \quad p_- = \text{Tr}(\rho\Pi_-) = |\beta|^2. \quad (32)$$

The complete real apparatus is more accurately represented by an instrument $\{\mathcal{I}_+, \mathcal{I}_-\}$, because it not only assigns probabilities but also records the conditional state and detector change.

9.7 Full semantic breakdown

Semantic stage	Physical component	Mathematical structure	What becomes real
Persist	Prepared atomic beam	$ \Psi_0\rangle$	A repeatable input condition.
Distinguish	Magnet axis and field gradient	$\{\Pi_+, \Pi_-\}$	The z -spin alternatives become the enacted question.
Relate	Magnetic coupling	$H_{\text{int}} = -\mu \cdot B$	Spin is linked to translational motion.
Propagate	Magnet plus free flight	U_{SM} and $ \varphi_{\pm}\rangle$	The distinction travels into spatial channels.
Resolve	Wavepacket separation and detector resolution	$ \langle\varphi_- \varphi_+\rangle \ll 1$	The channels become discriminable.
Recurse/Amplify	Detector, environment, electronics	$ D_{\pm}E_{\pm}\rangle$	The distinction is copied into many later relations.
Close	Stored spot and conditional state	$\{p_k, r_k, \rho_{\{S k\}}\}$	A retained observer-relative fact.

Semantic stage	Physical component	Mathematical structure	What becomes real
Reflect/Denote	Human scientific observer	$\Lambda_H(\mathcal{M},k)$	The observation machinery itself becomes an explicit model.

Worked-system conclusion

The observed spin result is not attributable to the atom alone. It is the closure of the atom's admissible state, the magnet's selected distinction, the spin-position coupling, free propagation, detector resolution, environmental stabilisation, and the retained record. Human observation adds a model of that entire machinery.

10. Semantic Compression of the Complete Operation

Once the conventional mathematics has been displayed, the entire Stern-Gerlach chain can be compressed without losing its denotation:

$$Q[\psi] \rightarrow D_n \rightarrow R_{\{\mu B\}} \rightarrow G_x \rightarrow V_D \rightarrow K_E \rightarrow C_k \rightarrow \Lambda_H. \quad (33)$$

The symbols denote

Class	Expanded meaning	Stern-Gerlach realisation
$Q[\psi]$	Coherent unresolved admissibility	Incoming spin state
D_n	Mechanical selection of distinction n	Magnet orientation
$R_{\{\mu B\}}$	Physical relation that couples alternatives to channels	Magnetic interaction
G_x	Propagation into an observable carrier	Spatial separation
V_D	Resolution by observer machinery	Detector discriminability
K_E	Recursive retention and environmental stabilisation	Pointer record plus decoherence
C_k	Closure into outcome k and conditional state	Stored upper or lower detection
Λ_H	Reflexive semantic denotation	Human report and mathematical model

This compression is valid only because every class has been denoted once against the expanded mechanics. It is therefore not a replacement for the mathematics but a reusable relational language for its physical necessity and order.

11. Consequences and Limits

11.1 Superposition is related to reality through observation machinery

Superposition is not a pre-reality fiction. It is the unresolved physical account retained before a mechanically stable distinction exists for a given observer. Mechanical observation maps that account into a family of possible records and conditional states:

$$\rho_S \xrightarrow{\mathcal{M}_O} \{p_k, r_k, \rho_{\{S|k\}}\}. \quad (34)$$

11.2 Reality is observer-relative without being arbitrary

A record is relative to an observer architecture, but it is not freely invented. The state, coupling, conservation laws, detector thresholds, and environmental dynamics constrain what records can occur. Relativity here means physically indexed, not subjective whim.

11.3 Mechanical observation does not require consciousness

The detector record can form before any human reads it. Human consciousness enters as a higher-order, reflexive, semantically expressive observer architecture. This preserves a physical account of measurement while giving human observation genuinely additional structure.

11.4 Decoherence is not silently equated with a complete interpretation

The paper uses decoherence to explain stable pointer records and effective loss of locally accessible interference. It does not assume that decoherence alone settles every question about unique global outcomes. The Principles of Observation and Mechanical Observation are intended as common operational machinery beneath competing interpretations, not as an unsupported declaration that all interpretive questions have vanished.

11.5 Testable use of the framework

The principles become scientifically productive when they expose a missing observer component, predict how changing observer architecture changes the measurable distinction, or prove that two apparently equivalent measurement descriptions do not preserve the same complete logical action. Candidate tests include imperfect detector resolution, weak measurement, delayed readout, measurement-chain reversibility, and quantum-reference-frame changes.

12. Formal Summary and Research Programme

PO - universal observation

A distinction becomes physically accountable relative to another system through relation, persistence, and closure.

$$\mathcal{D}_{\{O \leftarrow S\}} = \mathcal{C}_O \circ \mathcal{P}_O \circ \mathcal{R}_{\{OS\}} \circ \mathcal{D}_O(S).$$

PMO - structured observation

An observer architecture converts admissible system distinctions into stable records through prepared machinery, channel-defining coupling, propagation, resolution, amplification, and record production.

$$\mathcal{D}^{\text{mech}}_{\{O \leftarrow S\}} = \mathcal{C}_O \circ \mathcal{R}_{\text{ec}_O} \circ \mathcal{A}_O \circ \mathcal{P}_{\text{roj}_O} \circ \mathcal{U}_{\{SM\}}.$$

Human reflexive observation

A mechanical observation whose record and machinery can be recursively modelled and semantically denoted by the observer.

$$\mathfrak{D}_H^{\wedge(2)}(S) = \Lambda_H \circ \text{Rec}_H \circ \mathfrak{D}_{\{H \leftarrow S\}^{\wedge\{\text{mech}\}}}.$$

The complete proposed quantum-to-reality operation is

Admissible logical actions → coherent state → observer-selected distinction → system-machinery entanglement → propagated record channels → decoherent stabilisation → observer-relative fact → reflexive semantic model. (35)

The next mathematical work is to:

1. Classify observer architectures by the distinctions they can physically enact.
2. Define equivalence of observation machinery under the Principle of Equivalent Action.
3. Use the Principle of Logical Action to distinguish apparatuses that give the same probabilities but different disturbance, memory, topology, or causal order.
4. Develop a relational normal form for quantum instruments and measurement chains.
5. Apply the framework to continuous measurement, field detection, gravitational observation, and human sensory systems.
6. Test whether the observer hierarchy provides a common bridge between quantum measurement, relational state attribution, and classical record formation.

Final statement

Superposition is unresolved admissible reality. Observation is the production of a retained distinction relative to another system. Mechanical observation is the physical machinery that makes the distinction discriminable and persistent. Human scientific observation is the recursive capacity to make that machinery itself observable and mathematically denotable.

References

- [1] C. Rovelli, “Relational Quantum Mechanics,” *International Journal of Theoretical Physics* 35, 1637-1678 (1996); arXiv:quant-ph/9609002.
- [2] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Reviews of Modern Physics* 75, 715-775 (2003); arXiv:quant-ph/0105127.
- [3] M. Ozawa, “Quantum Measurement Theory for Systems with Finite Dimensional State Spaces,” arXiv:2110.03219 (2021); see also K. Okamura and M. Ozawa, “Measurement theory in local quantum physics,” *Journal of Mathematical Physics* 57, 015209 (2016); arXiv:1501.00239.
- [4] A. Reddy, J. Samuel, K. Shivam, and S. Sinha, “Coarse Quantum Measurement: An analysis of the Stern Gerlach experiment,” arXiv:1509.09024 (2015).
- [5] A. Venugopalan, D. Kumar, and R. Ghosh, “Analysis of the Stern-Gerlach Measurement,” arXiv:quant-ph/9501022 (1995).

Appendix A. Compact Notation Ledger

Symbol	Name	Operational definition
$\mathfrak{D}$	Observation	Retained observer-relative distinction.

Symbol	Name	Operational definition
$\mathfrak{D}^{\text{mech}}$	Mechanical observation	Structured state-to-record conversion by observer machinery.
$\mathcal{M}_0$	Observer architecture	Ready state, coupling, channels, amplification, and record structure.
$Q[\psi]$	Unresolved admissibility	Coherent account of available logical actions.
D_n	Mechanical distinction	Question physically selected by machinery orientation or basis.
R	Relation	Coupling through which alternatives become reachable channels.
G	Propagation	Transport of the distinction through physical degrees of freedom.
V	Resolution	Production of discriminable channels or records.
K	Recursive retention	Copying, amplification, environmental stabilisation, or memory.
C	Closure	Complete probability and conditional state account.
Λ_H	Human denotation	Reflexive model, language, diagram, or equation.